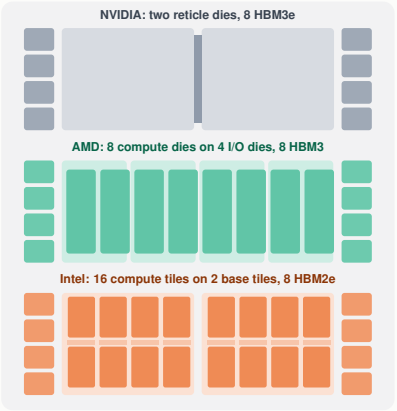
LECTURE NOTES · HARDWARE, 2026

# Data-Center GPU Platforms Compared

NVIDIA from the A100 to Vera Rubin, AMD Instinct from the MI100 to the MI455X and the Helios rack, and Intel from the Gen graphics families through Xe and Ponte Vecchio to what followed: the compute units, the packages, the memory, the fabrics, the racks and the software.

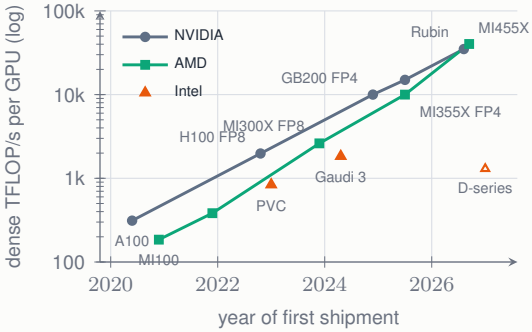



three answers to the reticle limit: Blackwell (2024), MI300X (2023), Ponte Vecchio (2023); HBM stacks at the sides

# Three vendors, one decade: the shape of the race

Per-GPU tensor peaks from the three vendors since 2020; the lines converge on paper, the market does not.

fastest dense tensor peak, vendor figures†




**3**
vendors; one at ∼80% share†

**5**
exascale systems, June 2026

**What the curves say**

**On paper the peaks converge:** AMD's MI355X matches the GB200's 10 PFLOP/s of dense FP4 and the MI455X exceeds Rubin's 35; Intel's Ponte Vecchio sat between A100 and H100.

**The market did not:** NVIDIA held about 80% of data-center AI accelerators in 2025 by secondary estimates, AMD under 10%, Intel under 1%†.

**HPC is the exception:** of the five June 2026 exascale systems, three run AMD or Intel GPUs, one NVIDIA, one no accelerator. The deck is about why both are true.

# How to compare a platform, and the decade's timeline

A GPU is sold as a package on a board in a rack with a fabric and a software stack; each level is a separate contest.

six levels, and the names each vendor gives them†

| level | NVIDIA | AMD | Intel |
|---|---|---|---|
| compute unit | SM; warp of 32 | CU; wavefront of 64 | Xe-core; subgroup 8–32 |
| package | GB200: 2 dies | MI300X: 8 XCD, 4 IOD | PVC: 47 tiles |
| board | SXM / HGX 8-GPU | OAM / UBB 8-GPU | OAM 4–8 GPU |
| scale-up fabric | NVLink, NVSwitch | Infinity Fabric | Xe Link |
| rack | NVL72 | Helios (72) | none shipped |
| software | CUDA | ROCm, HIP | oneAPI, SYCL |




### Why compare at every level

A faster die behind a slower fabric loses the training run; a better rack with a thinner library loses the deployment. The contests are decided separately, and the winners differ: Intel leads in packaging, AMD in memory per GPU and FP64, NVIDIA in the fabric and the software. The scorecard is slide 31.

### The decade in one sentence each

**NVIDIA** turned an architecture lead into a platform: fabric, CPU, rack and libraries. **AMD** built the GPUs of two of the first three exascale machines and, from 2023, a credible AI line that reached rack scale in 2026. **Intel** shipped one exascale GPU, cancelled its two successors, and is restarting from inference parts.

# The NVIDIA line as platforms: HGX A100 to Vera Rubin NVL72

Six platforms in six years: the board became a rack, memory grew 7×, the scale-up link 5×, FP64 went up and down.

| platform† | HGX A100 | HGX H100 | HGX H200 | GB200 NVL72 | GB300 NVL72 | VR NVL72 |
|---|---|---|---|---|---|---|
| first shipped | 2020 | 2022 | 2024 | 2024 | 2025 | 2026 |
| GPU | A100 | H100 | H200 | B200 | B300 | Rubin |
| HBM per GPU (GB) | 40 / 80 | 80 | 141 | 186 | 288 | 288 |
| bandwidth (TB/s) | 1.6 / 2.0 | 3.35 | 4.8 | 8.0 | 8.0 | 19.2 |
| FP64 (TF) | 9.7 / 19.5 | 34 / 67 | 34 / 67 | 40 | 1.4 | 33 |
| FP16 dense (TF) | 312 | 989 | 989 | 2,500 | 2,500 | 4,000 |
| FP8 dense (TF) | — | 1,979 | 1,979 | 5,000 | 5,000 | 17,500 |
| FP4 dense (TF) | — | — | — | 10,000 | 15,000 | 35,000 |
| scale-up, GB/s per GPU | 600 | 900 | 900 | 1,800 | 1,800 | 3,000 |
| GPUs per domain | 8 | 8 | 8 | 72 | 72 | 72 |
| CPU | x86 host | x86 host | x86 host | Grace | Grace | Vera |
| GPU power (W) | 400 | 700 | 700 | 1,200 | 1,400 | n.p. |
| board or rack (kW) | 6.5 | 10.2 | 10.2 | ∼120 | ∼130 | n.p. |

datasheet values, dense tensor peaks (NVIDIA headlines are the 2:4-sparse figures, twice these); FP64 as vector / tensor where they differ; power for the DGX A100 and DGX H100 systems and the NVL72 racks; Rubin values from the spec page revised in September 2026; n.p.: not published

**Reading the line**

**The unit grew:** an 8-GPU HGX board behind an x86 host became a 72-GPU rack with NVIDIA's own CPUs; the NVLink domain is the thing a customer buys.

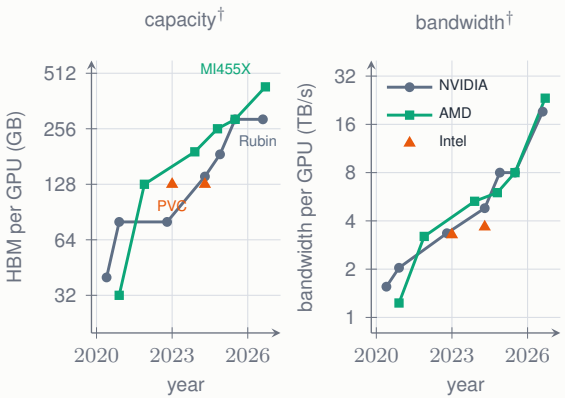
**Memory 7×, bandwidth 12×:** 40 GB at 1.6 TB/s to 288 GB at 19.2 TB/s per GPU; the H200 was a memory refresh of an unchanged die, the B300 a rebalanced one.

**Precision down, peaks up:** FP16 to FP8 to FP4; the dense peak rose 110× (derived). FP64 rose to 67 TFLOP/s on the H100's tensor cores, fell to 40 and then to 1.4, and came back to 33 on Rubin.

**What to compare against:** AMD matches this line GPU for GPU from 2020 (slide 7); Intel entered once, with Ponte Vecchio against the H100 (slide 17).

# Memory per GPU: the contest AMD kept winning on paper

HBM per GPU, 2020–2026: AMD led from the MI250X on and tied in 2025; Ponte Vecchio matched it in 2023.




**Generation by generation**†

**2020–22:** MI100 32 GB trailed the A100's 40/80; MI250X 128 GB from two dies with four stacks each took the lead.

**2023–24:** MI300X 192 GB at 5.3 TB/s against the H100's 80 at 3.35; the H200 (141 GB) and MI325X (256 GB) answered each other; Ponte Vecchio's 128 GB matched the MI250X in early 2023.

**2025:** both at 288 GB of 12-high HBM3e at 8 TB/s.

**2026:** MI455X 432 GB of HBM4 on 12 stacks at 23.3 TB/s against Rubin's 288 GB on 8 stacks at 19.2.

**Why it did not decide:** inference is bound by memory, so capacity per GPU sets how few GPUs a model needs; but a 72-GPU NVLink rack offsets a per-GPU deficit, and HBM supply from three makers is everyone's bottleneck.

# Compute peaks per GPU: three vendors, FP64 to FP4

Dense vendor peaks; AMD doubled FP64 on matrix cores until CDNA 4; Intel's PVC ran FP64 at full rate.

| GPU† | year | process | FP64 vec / mat | FP16 | FP8 | FP4 | W |
|---|---|---|---|---|---|---|---|
| A100 | 2020 | N7 | 9.7 / 19.5 | 312 | — | — | 400 |
| H100 SXM | 2022 | 4N | 34 / 67 | 989 | 1,979 | — | 700 |
| GB200 | 2024 | 4NP | 40 / 40 | 2,500 | 5,000 | 10,000 | 1,200 |
| GB300 | 2025 | 4NP | 1.4 | 2,500 | 5,000 | 15,000 | 1,400 |
| Rubin | 2026 | 3 nm | 33 / 33 | 4,000 | 17,500 | 35,000 | n.p. |
| MI100 | 2020 | N7 | 11.5 / — | 185 | — | — | 300 |
| MI250X | 2021 | N6 | 47.9 / 95.7 | 383 | — | — | 560 |
| MI300X | 2023 | N5 + N6 | 81.7 / 163 | 1,307 | 2,615 | — | 750 |
| MI355X | 2025 | N3P + N6 | 78.6 / 78.6 | 2,500 | 5,000 | 10,000 | 1,400 |
| MI455X | 2026 | N2 + N3P | n.p. | 5,030 | 20,130 | 40,260 | n.p. |
| PVC 1550 | 2023 | N5 + Intel 7 | 52.4 / 52.4 | 839 | — | — | 600 |
| Gaudi 3 | 2024 | 5 nm | — | 1,835 | 1,835 | — | 900 |

TFLOP/s, dense, matrix units for FP16 and below (MI100 FP16 is 185 FP16 / 92 BF16; MI455X figures are MXFP8 and MXFP4); PVC compute tiles N5, base tiles Intel 7, link tiles N7; PVC FP16 is the vendor figure, consistent with its confirmed 419 TF TF32; Gaudi 3 reports one figure for BF16 and FP8; Rubin from the September 2026 spec page

**How to read it**

**Low precision converged:** 10 PFLOP/s of dense FP4 on both the GB200 and the MI355X in 2025; the MI455X's 40 against Rubin's 35 in 2026. Peaks are a function of process, die area and power, and all three vendors buy from TSMC.

**FP64 diverged:** AMD kept matrix FP64 at 2× the vector rate from MI250X to MI300X and reached 163 TFLOP/s, halved the matrix rate on CDNA 4, and promises 288 on the HPC part MI430X (2027). NVIDIA peaked at 67 and cut to 1.4 on the B300 (slide 12).

**Intel's bet:** Ponte Vecchio ran FP64 at the full FP32 rate, 52 TFLOP/s, more than the H100's vector rate in 2023, with a tensor unit a generation behind.

**Peak is not achieved:** fractions of 45–75% on matrix multiply (slide 26).

# AMD Instinct and CDNA: the compute unit, five generations

A GCN-derived CU with 64-wide wavefronts (32 on CDNA 5) and matrix cores, from one die to a dozen chiplets.

one CDNA 3 compute unit (AMD whitepaper, 2023)†




a 64-work-item wavefront issues on a 16-lane SIMD over four cycles; four SIMDs give 64 FP32 lanes (128 FLOP per clock, 256 in CDNA 3) plus matrix cores at 2–16× that; compare the SM's four 32-lane partitions

| CDNA† | 1: MI100 | 2: MI250X | 3: MI300X | 4: MI355X | 5: MI455X |
|---|---|---|---|---|---|
| shipped | Nov 2020 | Nov 2021 | Dec 2023 | Q3 2025 | Q3 2026 |
| compute dies | 1 (N7) | 2 GCD (N6) | 8 XCD (N5) | 8 XCD (N3P) | 8 XCD (N2) |
| base dies | — | — | 4 IOD (N6) | 2 IOD (N6) | 2 IOD, 2 FCD |
| transistors | 25.6B | 58.2B | 153B | 185B | 320B |
| CUs | 120 | 220 | 304 | 256 | 256 WGP |
| HBM: GB, TB/s | 32, 1.2 | 128, 3.2 | 192, 5.3 | 288, 8.0 | 432, 23.3 |
| power (W) | 300 | 560 | 750 | 1,400 | n.p. |

**The lineage, in one paragraph each†**

**GCN to CDNA (2020):** AMD split its GPU line into RDNA for graphics and CDNA for compute. The MI100's CU kept GCN's 64-wide wavefront and added matrix cores; 11.5 TFLOP/s FP64, PCIe card.

**CDNA 2 (2021):** two dies in one OAM module, FP64 at the FP32 rate, packed FP32, 128 GB of HBM2e; the Frontier GPU (slide 8).

**CDNA 3 (2023):** compute dies stacked on I/O dies that carry a 256 MB Infinity Cache; a GPU (MI300X) and an APU with Zen 4 cores (MI300A, El Capitan; slide 9).

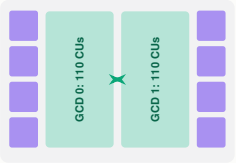
**CDNA 4 (2025):** N3P compute dies, 288 GB, FP4 and FP6, matrix FP64 halved (slide 10).

**CDNA 5 (2026):** N2 compute dies, work-group processors with 32-wide waves, HBM4, UALink over Ethernet, a 72-GPU rack (slide 11).

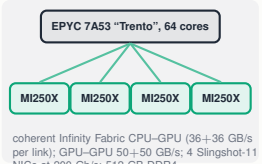
# MI250X and Frontier: two dies per module and the first exaflop

Each MI250X holds two compute dies that software sees as two GPUs; 37,632 of them made Frontier (2022).

an MI250X module (left) and a Frontier node (right)†




in-package Infinity Fabric 400 GB/s; 4 HBM2e stacks per GCD, 64 GB each; 8 external IF links




coherent Infinity Fabric CPU–GPU (36+36 GB/s per link); GPU–GPU 50+50 GB/s; 4 Slingshot-11 NICs at 200 Gb/s; 512 GB DDR4

## The MI250X, by the numbers†

220 CUs, 14,080 lanes, 880 matrix cores at 1.7 GHz: 47.9 TFLOP/s FP64 and FP32, 95.7 FP64 matrix and packed FP32, 383 FP16; 128 GB HBM2e at 3.2 TB/s; 58.2B transistors (launch slide); 500–560 W OAM. Each die is a GPU to ROCm: a node of four modules is eight devices.

## Frontier and its siblings†

**Frontier (ORNL):** HPE Cray EX235a, 9,408 nodes of one Trento CPU and four MI250X: 37,632 GPUs, 75,264 dies, 74 cabinets, Slingshot-11 dragonfly. 1.102 EFLOP/s on HPL in June 2022, the first exascale result; 1.353 since November 2024, #3 in June 2026; 24.6 MW.

**The same design elsewhere:** LUMI (11,912 MI250X, 380 PFLOP/s), Setonix, Adastra, HPC6 (Eni, 478 PFLOP/s).

**Why AMD won these:** the 2019 CORAL-2 awards bought FP64 per dollar and per watt, a coherent CPU–GPU link from one vendor and an open stack (slide 12).

**The cost:** two GPUs per module and ROCm’s maturity in 2022 made the ports an effort of their own (slide 25).

# MI300X and MI300A: 3D chiplets and the exascale APU

Eight compute dies stacked on four I/O dies with a 256 MB cache; with CPU cores it became El Capitan.

the MI300X package: 4 IODs (N6) below, 8 XCDs (N5) above, 8 HBM3 stacks around†




XCDs are hybrid-bonded onto IODs that hold the 256 MB Infinity Cache (17.2 TB/s), the memory controllers and the Infinity Fabric; MI300A swaps two XCDs for three Zen 4 dies (24 cores) on the same 128 GB

**153B**

transistors, 12 dies

**5.3 TB/s**

192 GB HBM3, 8 stacks

**MI300X, MI300A, and their machines†**

**MI300X (December 6, 2023):** 304 CUs, 19,456 lanes, 1,216 matrix cores at 2.1 GHz: 81.7 TFLOP/s FP64 vector, 163.4 FP64 matrix and FP32, 1,307 FP16 dense, 2,615 FP8; 192 GB at 5.3 TB/s; 7 Infinity Fabric links of 128 GB/s on an 8-GPU UBB 2.0 board; 750 W: 2.4× the H100's memory, 1.3× its FP8.

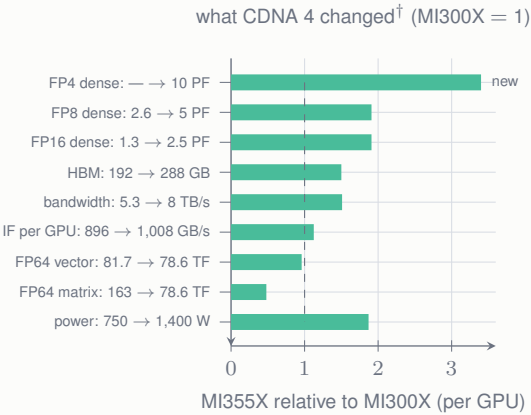
**MI300A:** 228 CUs and 24 Zen 4 cores sharing 128 GB of HBM3, 61.3 / 122.6 TFLOP/s FP64, 550–760 W.

**El Capitan (LLNL):** about 44,500 MI300A in nodes of four; 1.742 EFLOP/s in November 2024 (#1), 1.809 from November 2025, #2 in June 2026; 29.7 MW.

**MI325X (October 2024):** 256 GB of HBM3e at 6.0 TB/s and 1,000 W on the same silicon: the answer to the H200.

# CDNA 4: MI355X, 288 GB and FP4, matrix FP64 halved

The same recipe as Blackwell Ultra one quarter later: more HBM, 4-bit formats, less double precision, 1,400 W.




**CDNA 4 (announced June 12, 2025; shipping Q3 2025)†**

**Silicon:** 8 XCDs on TSMC N3P (32 of 36 CUs each, 256 in all) on 2 IODs on N6, 185B transistors, 2.4 GHz; LDS 160 KB per CU; 288 GB of 12-high HBM3e at 8 TB/s; Infinity Fabric 7 × 144 GB/s; 1,400 W liquid (MI350X 1,000 W air).

**Formats:** MXFP6 and MXFP4 at 10 PFLOP/s dense (no sparse figure on the datasheet; marketing says 20), FP8 5, FP16 2.5: the GB200's numbers. Matrix FP64 per CU per clock halved, so FP64 is 78.6 TFLOP/s vector and matrix alike: half the MI300X's matrix rate, 56× the B300's.

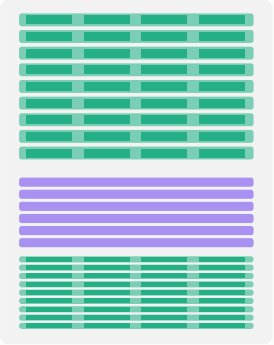
**Measured:** first MI355X and MXFP4 results in MLPerf Inference 5.1 (Sep 2025); in 6.0 (Apr 2026) 93–119% of the B200 on Llama 2 70B (slide 30).

**Deployed:** ORNL's Lux AI cluster (early 2026).

# MI400, MI455X and Helios: AMD's rack (2026)

CDNA 5 on TSMC N2, 432 GB of HBM4, UALink over Ethernet, 72 GPUs per rack; shipping from Q3 2026.

Helios: 72 MI455X, 18 EPYC "Venice" CPUs, 18 DPUs, 800G NICs, in one OCP Open Rack Wide†




## MI455X and the deals†

**The GPU (July 23, 2026):** 8 XCDs on TSMC N2 with 2 cache and 2 I/O dies on N3P, 320B transistors; 256 work-group processors; 192 MB of shared L2, no Infinity Cache; 432 GB of HBM4 on 12 stacks at 23.3 TB/s (19.6 announced in 2025); 40.3 PFLOP/s MXFP4, 20.1 MXFP8, 5.0 FP16, 315 TFLOP/s FP32 dense; FP64 and power not published; 3.6 TB/s UALoE per GPU.

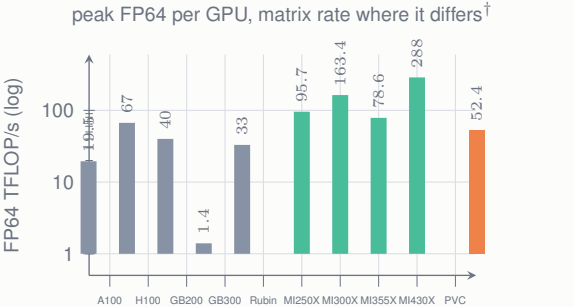
**Status:** MI450 series launched in Q3 2026; Helios racks running at Anthropic, Meta, Microsoft, OpenAI and Oracle (AMD 8-K, November 3).

**Commitments:** OpenAI 6 GW with a warrant for up to 160M shares; Oracle about 50,000 MI450; Meta up to 6 GW; Anthropic up to 2 GW.

**Next:** MI430X for HPC (288 TFLOP/s FP64, H1 2027); MI500 in 2027, MI600 in 2028.

# FP64 across the three vendors: who still builds for HPC

AMD kept doubling it, NVIDIA cut it, Intel ran it at full rate once; three of five exascale machines are AMD or Intel.




MI430X is a 2027 part with a vendor-stated figure; NVIDIA's Blackwell and Rubin have no separate matrix FP64; PVC's FP64 equals its FP32 rate

**Why the lines diverged†**

**AMD designed to contracts:** the 2019 CORAL-2 awards (Frontier, El Capitan) and Europe's procurements specified FP64; MI250X and MI300A are those machines' GPUs, and the MI430X is sold as a "true hardware FP64" part for Discovery (ORNL, 2028), Alice Recoque and LUMI-AI.

**NVIDIA designed to inference:** the B300 removed most FP64 units; Rubin restored 33 TFLOP/s, half the H100's tensor rate; emulation on INT8 tensor cores is offered instead.

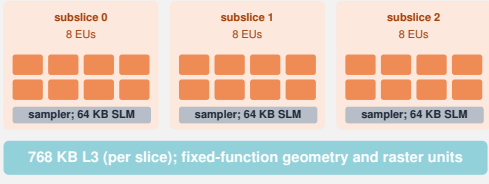
**Intel's one entry:** Ponte Vecchio's full-rate FP64 gave Aurora its exaflop; its successors were cancelled.

**The scoreboard:** El Capitan, Frontier, Aurora, JUPITER, LineShine. Ask: vector or matrix rate, native or emulated, and the fraction achieved on your own codes (slide 26).

# Intel's integrated GPUs: Gen9 to Gen12, the EU and the slice

For a decade Intel's GPU shipped inside every laptop CPU; its EU and slice are what Xe grew from.

a Gen9 slice (Skylake, 2015): 3 subslices of 8 execution units; Junkins, Intel whitepaper†




an EU: 7 hardware threads, each with 128 registers of 32 bytes, two SIMD-4 FPUs issuing per clock: 16 FP32 FLOP per clock per EU; GT2 has one slice (24 EUs, 384 FLOP/clk), GT4 three (72 EUs); SF: SuperFin

| generation† | Gen9 | Gen11 | Xe-LP (Gen12) | Xe-LP DG1 |
|---|---|---|---|---|
| product | Skylake | Ice Lake | Tiger Lake | Iris Xe MAX |
| year, process | 2015, 14 nm | 2019, 10 nm | 2020, 10 nm SF | 2020, 10 nm SF |
| EUs (GT2 or full) | 24 | 64 | 96 | 96 |
| FLOP per clock | 384 | 1,024 | 1,536 | 1,536 |
| FP32 TFLOP/s | about 0.4 | about 1 | about 2.1 | about 2.5 |

**What the integrated years built†**

**The EU:** a small multithreaded core, seven threads deep to hide latency, with two four-wide FPUs; a shader's SIMD width (8, 16 or 32) is chosen by the compiler, not fixed by hardware. Compare the 32-thread warp and the 64-wide wavefront.

**The hierarchy:** EU, subslice (8 EUs, a sampler, 64 KB of shared local memory), slice (3 subslices, 768 KB L3), one to three slices per GPU; Xe renamed every level (slide 14).

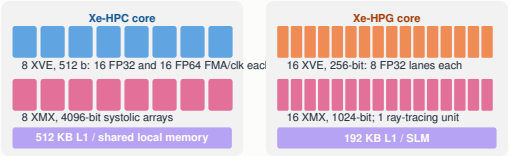
**Scale:** 384 FLOP per clock on Gen9's 24 EUs; about 1 TFLOP/s on Gen11's 64; about 2 TFLOP/s on Xe-LP's 96 EUs (derived). The DG1 (2020) was Intel's first discrete GPU since the i740 of 1998.

**The software:** OpenCL since 2011, then oneAPI (2018) with Level Zero (2020); hundreds of millions of iGPUs were the argument for a data-center line on the same design.

# From EU to Xe-core: the Xe hierarchy against SM and CU

Xe (2020) renamed the units, doubled the vector width for HPC and added the XMX matrix engine.

an Xe-HPC core (Ponte Vecchio) beside an Xe-HPG core (Arc)†




| level† | NVIDIA | AMD CDNA | Intel Xe-HPC |
|---|---|---|---|
| thread group | warp, 32 | wavefront, 64 | subgroup, 8–32 |
| vector unit | partition: 32 lanes (Hopper) | SIMD: 16 lanes, 4 cycles | XVE: 16 lanes |
| core | SM: 4 partitions, 4 tensor cores | CU: 4 SIMDs, 4 matrix cores | Xe-core: 8 XVE, 8 XMX |
| cluster | GPC: 18 SMs | XCD: 38–40 CUs | Xe-slice: 16 cores |
| die | 132 SMs (H100); 148 on B200's two | 304 CUs (8 XCDs) | stack: 64 cores |

**The renaming, and what changed underneath†**

**Names:** EU became Xe Vector Engine (XVE); subslice became Xe-core; slice became render slice (HPG) or Xe-slice (HPC); a stack is a die with four slices, four HBM controllers and eight Xe Links.

**Two variants:** Xe-HPG (Arc) keeps 256-bit XVEs and adds ray-tracing units; Xe-HPC doubles the XVE to 512 bits, runs FP64 at the FP32 rate, quadruples the XMX to 4096 bits, and gives each core 512 KB of L1/SLM, twice an H100 SM.

**XMX:** systolic arrays for INT8, FP16/BF16 and TF32, the counterpart of tensor and matrix cores, with no FP64 mode.

**Across vendors:** an Xe-core, an SM and a CU are the same kind of object; lane counts and thread-group width differ, and Intel's SIMD width is a compile-time choice.

# Xe-HPG, Xe2, Xe3: Arc Alchemist to Panther Lake

Three discrete generations in four years; Intel's data-center restart is built from this line.

| product† | Arc A770 | Arc B580 | Arc Pro B60 | Panther Lake |
|---|---|---|---|---|
| architecture | Xe-HPG | Xe2 | Xe2 | Xe3 |
| launched | Oct 2022 | Dec 2024 | 2025 | Jan 2026 |
| die | ACM-G10 | BMG-G21 | BMG-G21 | (integrated) |
| process | TSMC N6 | TSMC N5 | TSMC N5 | Intel 18A |
| transistors (B) | 21.7 | 19.6 | 19.6 | — |
| die area ($mm^2$) | 406 | 272 | 272 | — |
| Xe-cores | 32 | 20 | 20 | up to 12 |
| FP32 lanes | 4,096 | 2,560 | 2,560 | 1,536 |
| XMX engines | 512 | 160 | 160 | 96 |
| FP32 TFLOP/s | 17.2 | 14.6 | — | — |
| INT8 TOPS (XMX) | — | 233 | 197 | about 120 |
| memory (GB) | 16 GDDR6 | 12 GDDR6 | 24 GDDR6 | LPDDR5X |
| bandwidth (GB/s) | 560 | 456 | 456 | — |
| power (W) | 225 | 190 | 120–200 | 25 |

A770 XMX FP16 peak 138 TFLOP/s; Xe2 moved the XVE to SIMD16 and halved the XMX count per core; Panther Lake's 12 Xe3 cores at 2.5 GHz give about 120 TOPS on the GPU plus about 50 on the NPU; Lunar Lake (Sept 2024) was the first Xe2 part

**Three generations, and the restart†**

**Alchemist (2022):** the first high-end discrete Intel GPU since 1998: 32 Xe-cores, ray tracing, XMX, 406 $mm^2$ on TSMC N6, $329; drivers were the problem for a year.

**Battlemage (2024):** Xe2 on TSMC N5, fewer and wider cores (SIMD16), 272 $mm^2$, $249; the Arc Pro B50/B60 (2025) put 16–24 GB on workstation cards, and "Project Battlematrix" stacked eight B60s (192 GB) for local inference.

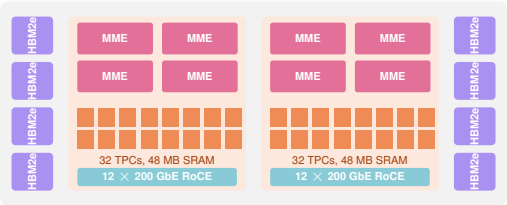
**Celestial (Xe3, 2026):** first as Panther Lake's integrated GPU on Intel 18A; the discrete Xe3P is the basis of the GPU D-series (slide 20).

**Why it matters here:** after Falcon Shores, Intel's data-center GPU plan is built from this graphics line, as NVIDIA's L40S is from Ada.

# Gaudi 2 and Gaudi 3: Intel's other accelerator

The Habana architecture: matrix engines, tensor cores and Ethernet on the die; strong on paper, weak in the market.

Gaudi 3 (2024): two dies, 8 MMEs, 64 TPCs, 24 Ethernet ports, 8 HBM2e stacks†




MME: matrix multiplication engine (a large systolic array); TPC: tensor processor core, a VLIW SIMD engine for the rest; scale-up and scale-out both over standard Ethernet from the chip, no proprietary fabric

**1,835 TF**

BF16 and FP8, dense

**128 GB**

HBM2e, 3.67 TB/s

**The line and its outcome†**

**Origin:** Habana Labs, bought by Intel in December 2019 for about $2B; Gaudi (2019) targeted training with Ethernet on the chip.

**Gaudi 2 (May 2022):** 7 nm, 24 TPCs, 96 GB HBM2e at 2.45 TB/s, 24 × 100 GbE, 600 W; in 2023 the only merchant non-NVIDIA accelerator with competitive MLPerf Training results.

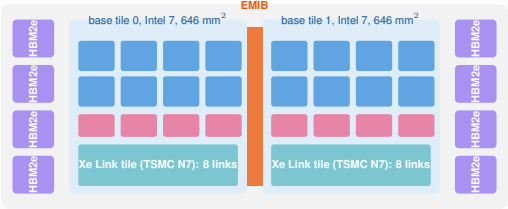
**Gaudi 3 (April 2024):** 5 nm, two dies, 64 TPCs and 8 MMEs, 128 GB HBM2e at 3.67 TB/s, 1,835 TFLOP/s BF16 or FP8, 24 × 200 GbE, 900 W air or 1,200 W liquid; no FP64, no FP4.

**Market:** Intel withdrew a $500M 2024 target in October 2024; analysts put its share below 1%; Intel's Q3 2026 results mention only the GPU D-series. A second ISA and stack (SynapseAI) divided Intel's software effort.

# Ponte Vecchio: 47 active tiles from five process nodes

Intel's exascale GPU (announced 2021, launched 2023) is the most complex package any vendor has shipped.

the Data Center GPU Max 1550: two stacks on two base tiles, bridged by EMIB†




per stack: 8 compute tiles (TSMC N5, purple), 4 RAMBO cache tiles (Intel 7, orange), 1 Xe Link tile, 4 HBM2e stacks; 11 EMIB bridges; 47 active tiles, over 100 billion transistors, 2,330 mm² of active silicon

**128**
Xe-cores; 1,024 XVE, 1,024 XMX

**3.28 TB/s**
128 GB HBM2e, 8 stacks

## What the package is†

**Foveros and EMIB:** compute, cache and link tiles are stacked face-to-face on two large base tiles (Foveros, 3D), and the halves and the HBM are joined by embedded silicon bridges (EMIB, 2.5D): five process nodes from two foundries, two years before Blackwell's two dies.

**The cache:** 408 MB of L2 across base and RAMBO tiles, eight times the H100's 50 MB (the split is reported inconsistently); 64 MB of L1 and 64 MB of registers in all.

**Dates:** Architecture Day, August 2021; ISSCC (Gomes et al.), February 2022; Hot Chips 34; Max series announced November 2022, launched January 10, 2023, a year behind plan.

**SKUs:** Max 1550 (128 cores, 128 GB, 600 W OAM); Max 1100 (56 cores, 48 GB, 300 W PCIe); the Max 1350 was dropped.

# The Xe-HPC core; the Max 1550 against its contemporaries

Full-rate FP64, a 512 KB local store per core, a matrix engine a generation behind.

| GPU, 2022–23† | Max 1550 | H100 SXM | MI250X |
|---|---|---|---|
| shipped | Jan 2023 | Q4 2022 | Nov 2021 |
| process | N5 + Intel 7 + N7 | TSMC 4N | TSMC N6 |
| transistors | over 100B | 80B | 58.2B |
| cores | 128 Xe-cores | 132 SMs | 220 CUs |
| FP32 lanes | 16,384 | 16,896 | 14,080 |
| clock (GHz) | 1.6 | 1.98 boost | 1.7 |
| FP64 vector (TF) | 52.4 | 34 | 47.9 |
| FP64 matrix (TF) | — | 67 | 95.7 |
| FP32 (TF) | 52.4 | 67 | 47.9 |
| TF32 matrix (TF) | 419 | 494 | — |
| FP16 / BF16 matrix (TF) | 839 | 989 | 383 |
| INT8 (TOPS) | 1,678 | 1,979 | 383 |
| L2 | 408 MB | 50 MB | 2 × 8 MB |
| HBM | 128 GB, 3.28 TB/s | 80 GB, 3.35 TB/s | 128 GB, 3.2 TB/s |
| scale-up | 16 Xe Links | NVLink 4, 900 GB/s | 8 IF links, 800 GB/s |
| power (W) | 600 | 700 | 560 |

dense figures; Max 1550 FP16 and INT8 are Intel's figures and follow from its per-core table (4,096 FP16 and 8,192 INT8 ops per clock); H100 tensor peaks imply about 1.83 GHz; Xe Link runs 53 Gb/s per lane, lanes per link not published; the MI250X's two dies appear as two GPUs

**The core, and the arithmetic†**

**Lanes:** 128 cores × 8 XVE × 16 FP32 lanes = 16,384; × 2 FLOP × 1.6 GHz = 52.4 TFLOP/s, and the same for FP64 (Intel rates each XVE at 16 FP64 FMAs per clock); the H100's FP64 is half its FP32.

**XMX:** 419 TFLOP/s TF32 = 1,024 engines × 128 FMA per clock × 2 × 1.6 GHz (derived); BF16 doubles it, INT8 again. No FP8: FP8 work runs as FP16, 839 vs the H100's 1,979 TF (42%); INT8 near 2 POPS on both.

**Memory:** 512 KB of L1/SLM per core and 408 MB of L2 are the largest on-chip stores of the three; HBM bandwidth matches the H100.

**Verdict in 2023:** competitive FP64 and bandwidth; a generation behind on low-precision matrix work; 600 W; late.

# Aurora: 63,744 GPUs, 1.012 exaflops, 38.7 MW

Planned in 2015 for 2018; delivered in 2023 with Ponte Vecchio; exascale HPL on Intel GPUs in 2024.

an Aurora node: two Xeon Max CPUs with HBM, six Max 1550 GPUs, eight Slingshot NICs†




Xe Link between the six GPUs (all-to-all); PCIe 5 to the CPUs, 512 GB DDR5 per CPU; 8 Slingshot-11 NICs at 200 Gb/s per node; 10,624 nodes in 166 HPE Cray EX racks; 63,744 GPUs

| TOP500 entry† | Nov 2023 | Jun 2024 | Nov 2024–25 | Jun 2026 |
|---|---|---|---|---|
| nodes in the run | 5,439 | 9,234 | 9,234 | 9,234 |
| HPL (EFLOP/s) | 0.585 | 1.012 | 1.012 | 1.012 |
| rank | 2 | 2 | 3 | 4 |
| HPL-MxP (EF) | — | 10.6 | 11.6 | #3 |
| power (MW) | 24.7 | 38.7 | 38.7 | 38.7 |

**The machine, its history, and its science†**

**Peak:** 1.980 EFLOP/s FP64; HPL 1.012 EFLOP/s on 9,234 nodes, 51% of peak, at 38.7 MW: the least efficient of the exascale systems (Frontier 1.353 at 24.6 MW, El Capitan 1.809 at 29.7).

**History:** contracted in 2015 as a 180–200 PFLOP/s Knights Hill system for 2018; re-planned in 2017 as exascale for 2021; 7 nm slip 2020; installed June 2023; full HPL 2024.

**Science:** Argonne teams on four 2025 Gordon Bell finalists, two run on Aurora: AERIS, a weather and climate model at 10.21 EFLOP/s sustained in mixed precision on 10,080 nodes, and a light-matter dynamics code at 1.87 EFLOP/s with over a trillion atoms.

**Other PVC systems:** Dawn (Cambridge, 1,024 GPUs) and SuperMUC-NG Phase 2 (LRZ, 960).

# After Ponte Vecchio: cancellations and a restart from Xe3

Rialto Bridge and Falcon Shores were cancelled; Jaguar Shores has no date; the GPU D-series is sampling.

| product† | announced | what it was to be | what happened |
|---|---|---|---|
| Rialto Bridge | 2022 | Ponte Vecchio's successor: more Xe-cores, 800 W OAM, planned for 2023 | cancelled March 2023 in a "streamlined and simplified" roadmap (McVeigh) |
| Falcon Shores | 2022 | first an XPU (CPU+GPU tiles), then a GPU-only part for 2025 with Gaudi's networking | cancelled as a product January 30, 2025; "internal test chip" (Holthaus) |
| Gaudi 3 | Apr 2024 | Habana's training and inference accelerator, Ethernet-native | shipped; sales below the $500M 2024 target; to be folded into Jaguar Shores |
| Jaguar Shores | 2025 | "rack-scale" AI system successor to Falcon Shores and Gaudi | no date; not mentioned in Intel's Q3 2026 results |
| GPU D-series (Crescent Island) | Oct 2025 | inference GPU on Xe3P: 32 cores, 256 XMX, 1.3 PFLOP/s FP8, 160 GB LPDDR5X (480 GB on partner cards), 1.5 TB/s, 350 W PCIe | silicon at Hot Chips (Aug 2026), specified at OCP (Oct 2026), sampling; availability not stated |

### Why the data-center GPU faltered†

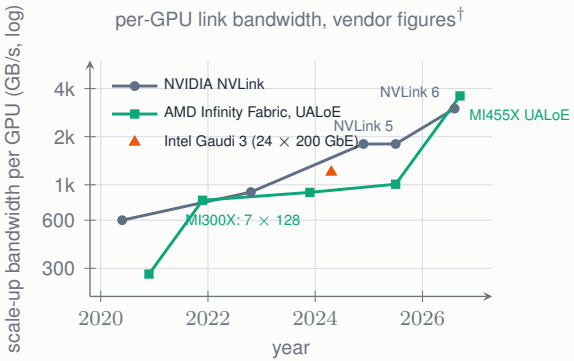
The 7 nm delay of 2020 pushed Ponte Vecchio a year late into Hopper's market; 47-tile packaging cost yield and power; oneAPI was young against CUDA; and the market moved from FP64 to inference, where PVC was weakest.

### The restart, and the NVIDIA deal†

The D-series uses LPDDR5X: more capacity at lower bandwidth and cost, for decode-bound inference in air-cooled servers. In September 2025 NVIDIA invested $5B in Intel; Intel will build x86 CPUs with NVLink. Intel's Q3 2026 data-center revenue grew 78%, on CPUs.

# Scale-up fabrics: NVLink, Infinity Fabric, Xe Link, UALink

The link between GPUs is NVIDIA's widest lead; AMD answered with a standard, Intel with Xe Link.




**Four fabrics†**

**NVLink (2016–):** switched by NVSwitch since 2018, so every GPU reaches every other at full rate; 600, 900, 1,800, 3,000 GB/s per GPU; in-switch reductions since Hopper; a 72-GPU domain since 2024.

**Infinity Fabric (xGMI):** direct links on an 8-GPU board, 7 × 128 GB/s on the MI300X; no switch, so a pair shares one link while NVLink spreads traffic over all: SemiAnalysis measured about 64 GB/s per pair against 450 in 2024.

**Xe Link:** PVC's 16 links join six GPUs in an Aurora node; never scaled to a switch.

**UALink (2024–):** the open standard (AMD, Intel, AWS, Google, Meta, Microsoft), 200 Gb/s per lane, up to 1,024 accelerators; Helios runs it over Ethernet: 3.6 TB/s per GPU.

# Rack-scale: the NVL72 family against Helios

Both sell 72-GPU liquid-cooled racks with their own CPUs; AMD's is wider, holds more HBM, shipped later.

| rack† | GB200 NVL72 | GB300 NVL72 | VR NVL72 | AMD Helios |
|---|---|---|---|---|
| first shipments | Nov 2024 | Jul 2025 | Aug 2026 | Q3 2026 |
| GPUs | 72 B200 | 72 B300 | 72 Rubin | 72 MI455X |
| CPUs | 36 Grace | 36 Grace | 36 Vera | 18 Venice |
| HBM total (TB) | 13.4 | 20 | 20.7 | 31 |
| HBM bandwidth (PB/s) | 0.58 | 0.58 | 1.4 | 1.7 |
| FP4 dense (EF) | 0.72 | 1.08 | 2.52 | 2.9 |
| FP8 dense (EF) | 0.36 | 0.36 | 1.26 | 1.4 |
| FP64 (PF) | 2.9 | 0.1 | 2.4 | n.p. |
| scale-up fabric | NVLink 5 | NVLink 5 | NVLink 6 | UALink (Eth.) |
| scale-up total (TB/s) | 130 | 130 | 216 | 260 |
| scale-out per GPU (Gb/s) | 400 | 800 | 1,600 | 2,400 |
| power (kW) | ∼120 | ∼130 | n.p. | 225–245 |
| form factor | 1 rack | 1 rack | 1 rack | 2 wide |

NVIDIA figures from product pages (Vera Rubin as revised September 2026); Helios from AMD's July 2026 launch and the Supermicro datasheet (which still lists 1.4 PB/s); Venice is Zen 6; GB200 FP8 dense derived from the sparse 720 PF; Helios scale-out: three 800G Vulcano NICs per GPU, 43 TB/s per rack; Helios power from press coverage

**Reading the two racks†**

**Same shape:** 72 GPUs, CPUs in the tray, 216–260 TB/s of scale-up in 2026, liquid cooling. AMD reached NVIDIA's form two years after the GB200 NVL72 with 1.5× the HBM and 1.15× the dense FP4 of Vera Rubin.

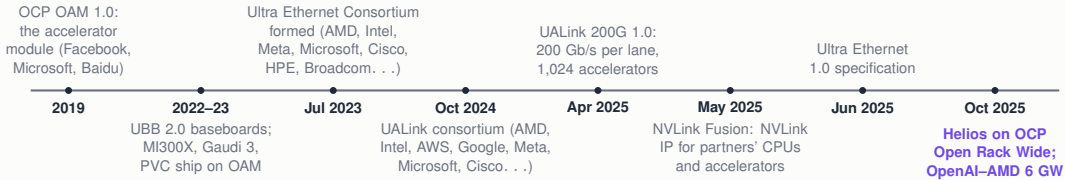
**Different fabric:** NVLink 6 is NVIDIA's own switch silicon with in-network compute; Helios runs UALink over merchant Ethernet switches: open, with Ethernet's latency.

**Different ratios:** one Grace or Vera per two GPUs against one Venice per four; one ConnectX-9 per GPU against three Vulcano 800G NICs; one rack against a double-wide one.

**Not yet comparable:** delivered racks and measured throughput (Helios shipping since Q3 2026).

# Standards: OAM, UBB, UALink, Ultra Ethernet, NVLink Fusion

Everyone but NVIDIA builds on open form factors and fabrics; NVIDIA opened its fabric to partners instead.




**The open stack, layer by layer**†

**Module and board:** OAM (102 × 165 mm, 700 W and later more) and the UBB 8-module baseboard let a server vendor build one chassis for AMD, Intel and others; SXM and HGX are NVIDIA's own. **Scale-up:** UALink, a memory-semantic fabric derived from Infinity Fabric. **Scale-out:** Ultra Ethernet, RDMA over Ethernet against InfiniBand.

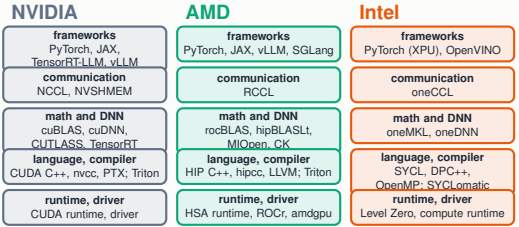
**What the standards have and have not done**†

They made the second source possible: an MI355X or Gaudi 3 drops into a UBB chassis, and Helios's fabric is bought from Ethernet switch vendors. They have not yet produced a shipped multi-vendor scale-up domain; UALink silicon was announced for 2026, and Helios is AMD-only. NVIDIA's answer, NVLink Fusion, admits a partner's chip to an NVLink domain on NVIDIA's terms.

# CUDA, ROCm and HIP, oneAPI and SYCL: three stacks

The same layers in each stack: driver, compiler, libraries, frameworks; what differs is age and breadth.

the three stacks, layer by layer (2026)†




portability layers sit above all three: Kokkos, RAJA, OpenMP offload, SYCL (Codeplay plugins for NVIDIA and AMD, 2022) and Triton

## Dates†

CUDA 1.0 June 2007. ROC(m) 1.0 April 2016, 6.0 December 2023, 7.0 September 2025, 10.0 August 2026 (8 and 9 skipped). oneAPI 1.0 September 2020; SYCL 2020 February 2021; UXL September 2023.

## What differs†

**Age and breadth:** CUDA has nineteen years of libraries and third-party code; ROCm began in 2016 and reached a stable cadence in 2025; oneAPI's first production toolkits shipped in December 2020.

**The porting path:** HIP is a CUDA-shaped API with `hipify` translating source, so most CUDA code compiles for AMD with small changes; SYCLomatic translates CUDA to SYCL (Intel claims 80–90% automatically).

**Frameworks:** PyTorch supports all three; vLLM and SGLang support NVIDIA and AMD; the kernels that set performance (FlashAttention, Triton) arrive first for CUDA.

**Intel's split:** oneAPI for Xe, SynapseAI for Gaudi.

**Governance:** CUDA is closed; ROCm is open source; oneAPI's specification sits in the Linux Foundation's UXL.

# Portability: what the SYCL and Kokkos studies measure

HPC codes run on all three vendors through portability layers; the gaps are in maturity, not the model.

| study† | what was run | finding |
|---|---|---|
| Reguly (SC-W 2023) | SYCL vs. native on Max 1100, A100, MI250X | DPC++ within 1.2% of CUDA on A100; 15.9% behind HIP on MI250X; weak on CPUs |
| Davis et al. (ICS 2025) | 7 models, 5 proxy apps, 5 GPUs | layers reach most of native speed; the spread is in compilers and runtimes |
| ORNL/TM-2025/4398 | Kokkos with SYCL back ends on AMD | DPC++ less mature on AMD than on Intel; HIP remains the reference |
| Aurora early science (2023–25) | SYCL, OpenMP, Kokkos ports of DOE codes | Summit and Frontier codes ran; tuning for the 512 KB SLM and two stacks took the time |

**The lesson of the studies**

The abstractions (parallel-for over work-groups, local memory, atomics) map to all three; what differs is compiler maturity, library coverage and vendor tuning. One source runs everywhere; it reaches peak where its author tuned.

**How HPC and AI differ here†**

**HPC codes are portable by construction:** DOE funded Kokkos, RAJA, OpenMP and SYCL ports for a decade because its machines use three vendors; Frontier and El Capitan (HIP), Aurora (SYCL, OpenMP) and the NVIDIA systems run the same applications.

**AI code is CUDA-first:** training and serving stacks are written to CUDA kernels and ported later; a port of FlashAttention to ROCm lags by months.

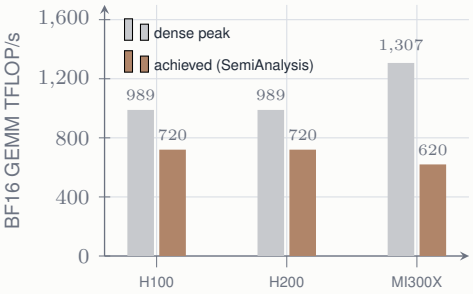
**The economics:** an HPC center buys for seven years and pays for the port once; an AI lab buys capacity by the quarter. This, more than any datasheet, is why the TOP500 is AMD and Intel while the AI clusters are NVIDIA.

**What changes it:** Triton, PyTorch compile and vLLM make the vendor a back end; AMD's 2026 deals assume that is far enough along.

# Achieved against peak: the software gap in numbers

In 2024 independent tests put MI300X matrix multiply at half its peak, the H100 at three quarters.

datasheet peak against measured, December 2024†




**What was measured, and what changed†**

**SemiAnalysis (December 2024):** on large BF16 matrix multiplies the H100 and H200 reached about 720 of 989 TFLOP/s (73%), the MI300X about 620 of 1,307 (47%); multi-node H100 training was 10–25% faster; GPU-to-GPU bandwidth measured about 64 GB/s per pair on Infinity Fabric against 450 on NVLink. “The CUDA moat is still alive”: competitive hardware, immature software.

**AMD’s response:** ROCm 6.3 to 7.0 rewrote the GEMM and attention libraries (hipBLASLt, Composable Kernel, AITER), a six-week cadence, and MLPerf submissions (slide 30).

**The rule:** a peak is a property of the die; the achieved fraction is a property of the libraries, and it rises over a product’s life. Compare measured throughput at the same date.

# The exascale systems: Frontier, Aurora, El Capitan, JUPITER

Four machines, three GPU vendors, three node designs; the FP64 side of the comparison, measured by HPL.

| system† | Frontier | Aurora | El Capitan | JUPITER |
|---|---|---|---|---|
| site | ORNL | Argonne | LLNL | Jülich |
| system | HPE Cray EX | HPE Cray EX | HPE Cray EX | BullSequana |
| node | 1 CPU + 4 MI250X | 2 CPU + 6 PVC | 4 MI300A | 4 GH200 |
| nodes | 9,408 | 10,624 | about 11,100 | about 6,000 |
| accelerators | 37,632 | 63,744 | about 44,500 | about 24,000 |
| first HPL (EF) | 1.102, 2022 | 0.585, 2023 | 1.742, 2024 | 0.793, 2025 |
| HPL, Jun 2026 | 1.353 EF (#3) | 1.012 EF (#4) | 1.809 EF (#2) | 1.000 EF (#5) |
| Rpeak (EF) | 2.06 | 1.98 | 2.82 | 1.23 |
| power (MW) | 24.6 | 38.7 | 29.7 | 15.8 |
| HPL per MW (PF) | 55 | 26 | 61 | 63 |
| network | Slingshot | Slingshot | Slingshot | InfiniBand |

TOP500 June 2026 entries; El Capitan's node count grew from 11,136 (Nov 2024) to 11,520 listed by LLNL; JUPITER's accelerator count from the FZJ press release and node count derived from four GH200 per node; Rpeak for Frontier and JUPITER from the list; HPL per MW derived

**Three node designs†**

**Frontier (2022):** one 64-core CPU for four GPUs over coherent Infinity Fabric; two dies per module, eight devices per node.

**Aurora (2023):** two 52-core CPUs with 64 GB of HBM each for six GPUs over PCIe 5; Xe Link between GPUs; eight NICs.

**El Capitan (2024):** no discrete CPU: four APUs of 24 Zen 4 cores and 228 CUs sharing 128 GB of HBM3 each; the highest HPL and best efficiency of the DOE machines.

**JUPITER (2025):** four Grace Hopper superchips per node (a 72-core CPU joined to an H100 at 900 GB/s); Europe's first exaflop.

**What HPL rewards:** FP64 matrix throughput and bandwidth per watt.

# TOP500 and Green500, June 2026: who runs what

A CPU-only Chinese system leads; AMD holds four of the top eight; the four most efficient are Grace Hopper.

| **TOP500, June 2026**† | | Rmax (PFLOP/s), accelerator, power (MW) | |
|---|---|---|---|
| 1 | LineShine, Shenzhen | CPU only (LX2) | 2,198 | 42.2 |
| 2 | El Capitan, LLNL | AMD MI300A | 1,809 | 29.7 |
| 3 | Frontier, ORNL | AMD MI250X | 1,353 | 24.6 |
| 4 | Aurora, Argonne | Intel Max 1550 | 1,012 | 38.7 |
| 5 | JUPITER, FZJ | NVIDIA GH200 | 1,000 | 15.8 |
| 6 | HPC7, Eni | AMD MI300A | 572 | 8.7 |
| 7 | Eagle, Microsoft Azure | NVIDIA H100 | 561 | — |
| 8 | HPC6, Eni | AMD MI250X | 478 | 8.5 |
| 9 | Fugaku, RIKEN | CPU only (A64FX) | 442 | 29.9 |
| 10 | Alps, CSCS | NVIDIA GH200 | 435 | 7.1 |

| **Green500, June 2026**† | | GFLOP/s per watt |
|---|---|---|
| 1 | KAIROS | GH200 | 73.3 |
| 2 | ROMEO-2025 | GH200 | 70.9 |
| 3 | Levante GPU extension | GH200 | 69.4 |
| 4 | Isambard-AI phase 1 | GH200 | 68.8 |
| 9 | Ouranos | MI300A | 66.5 |

**Reading the lists**†

**Five exascale systems,** three vendors of accelerator and one of none: LineShine's 2.198 EFLOP/s comes from Chinese CPUs, the first Chinese #1 since 2017.

**AMD:** four of the top eight (El Capitan, Frontier, HPC7, HPC6), all HPE Cray EX, all bought for FP64; the MI300A systems sit ninth and tenth on the Green500 at about 66 GFLOP/s per W against 73 for the best GH200 machine.

**Intel:** one system, Aurora, the least efficient of the exascale group; no PVC system added since 2023.

**NVIDIA:** JUPITER and Alps (GH200), Eagle (H100); the Green500's top four are Grace Hopper, then three H100 systems; no Blackwell or Rubin system is in the top ten.

**HPL-MxP:** El Capitan, Frontier, Aurora and LineShine lead, on their matrix units and software.

# The next machines and the gigawatt deals

DOE's 2026–28 systems split three ways; the AI commitments of 2025–26 are measured in gigawatts, not GPUs.

| HPC system† | site | accelerator | when |
|---|---|---|---|
| Doudna | NERSC | NVIDIA Vera Rubin (Dell) | late 2026 |
| Lux | ORNL | AMD MI355X (HPE) | early 2026 |
| Equinox, Solstice | Argonne (Oracle) | Blackwell, 10k / 100k | 2026–2028 |
| Discovery | ORNL | MI430X + Venice (HPE) | 2028 |
| Alice Recoque | CEA/GENCI | MI430X + Venice (Eviden) | 2026–27 |
| LUMI-AI | EuroHPC | AMD MI430X | H2 2027 |
| Blue Lion | LRZ | NVIDIA Vera Rubin (HPE) | 2027 |
| Herder | HLRS | AMD (after Hunter) | 2027 |

| AI commitment† | what | announced |
|---|---|---|
| OpenAI–AMD | 6 GW of Instinct from H2 2026; warrant up to 160M shares | Oct 2025 |
| Oracle–AMD | about 50,000 MI450 | Oct 2025 |
| Meta–AMD | custom MI450 part, up to 6 GW; warrant up to 160M shares | Feb 2026 |
| Anthropic–AMD | up to 2 GW of MI450 series; AMD invests up to $5B | Jul 2026 |
| Stargate Abilene | about 450,000 GB200, 1.2 GW | 2025 |
| NVIDIA–Intel | $5B stake; x86 CPUs with NVLink | Sep 2025 |

**What the procurements say†**

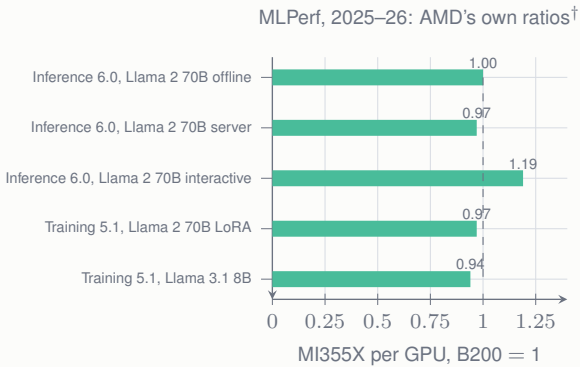
**DOE split the bet:** NERSC and Argonne's AI systems went to NVIDIA (Vera Rubin; Blackwell), ORNL's AI cluster and next leadership machine to AMD (MI355X; MI430X with Venice), Europe's follow-ons (Alice Recoque, LUMI-AI) to AMD's FP64 part. Intel has no announced next system.

**The AI deals are sized in power:** 6 GW is two to four million GPUs over several years, depending on the kW per GPU assumed (derived); AMD granted OpenAI and Meta warrants for 160M shares each, vesting with deployments; Oracle buys racks; AMD invests up to $5B in Anthropic. AMD's data-center revenue grew 97% in Q3 2026.

**Share:** secondary estimates still put NVIDIA at 75–80% of 2026 shipments; these deals are the first that could move it, and they are commitments, not yet deployments.

# MLPerf: where AMD and NVIDIA meet on the same benchmark

AMD entered in 2024 (inference) and 2025 (training); by 2026 the MI355X trades blows with the B200.




training ratios: B200 time over MI355X time (10.18 vs 9.85 min; 99.7 vs 93.7); inference ratios as AMD states them; against the B300: 0.92, 0.93, 1.04

**The rounds†**

**Inference 4.1 (August 2024):** AMD's first submission; eight MI300X within 2–3% of a DGX H100 on Llama 2 70B.

**Inference 5.0 (April 2025):** first MI325X results. **Training 5.0 (June 2025):** AMD's first training submission; the GB200's debut.

**Inference 5.1 (September 2025):** first MI355X and first MXFP4 results; Llama 2 70B at 2.7× the MI325X. **Training 5.1 (November 2025):** Llama 2 70B LoRA in 10.18 min against 9.85 (B200) and 9.59 (B300).

**Inference 6.0 (April 2026):** MI355X past one million tokens per second on 87 GPUs; the ratios at left.

**Training 6.0 (June 2026):** NVIDIA's 8,192-GPU GB300 runs set the scale; no Rubin or MI455X result yet.

**Intel:** Gaudi 2 submitted in 2023–24; no Gaudi 3 submission on record.

# The platforms in one table

Seven accelerators from three vendors, 2023–2026, as their datasheets state them.

| accelerator† | H100 | GB300 | Rubin | MI300X | MI355X | MI455X | PVC 1550 |
|---|---|---|---|---|---|---|---|
| vendor | NVIDIA | NVIDIA | NVIDIA | AMD | AMD | AMD | Intel |
| first shipped | 2022 | 2025 | 2026 | 2023 | 2025 | 2026 | 2023 |
| compute dies, process | 1, 4N | 2, 4NP | 2, 3 nm | 8 XCD, N5 | 8 XCD, N3P | 8 XCD, N2 | 16 tiles, N5 |
| transistors (B) | 80 | 208 | 336 | 153 | 185 | 320 | over 100 |
| HBM (GB) | 80 | 288 | 288 | 192 | 288 | 432 | 128 |
| bandwidth (TB/s) | 3.35 | 8.0 | 19.2 | 5.3 | 8.0 | 23.3 | 3.28 |
| on-chip cache | 50 MB L2 | n.p. | n.p. | 256 MB IC | 256 MB IC | 192 MB L2 | 408 MB L2 |
| FP64 vector / matrix (TF) | 34 / 67 | 1.4 | 33 | 81.7 / 163 | 78.6 | n.p. | 52.4 |
| FP16 dense (TF) | 989 | 2,500 | 4,000 | 1,307 | 2,500 | 5,030 | 839 |
| FP8 dense (TF) | 1,979 | 5,000 | 17,500 | 2,615 | 5,000 | 20,130 | — |
| FP4 dense (TF) | — | 15,000 | 35,000 | — | 10,000 | 40,260 | — |
| scale-up per GPU (GB/s) | 900 | 1,800 | 3,000 | 896 | 1,008 | 3,600 | Xe Link |
| scale-up domain | 8 | 72 | 72 | 8 | 8 | 72 | 6 (8 max) |
| power (W) | 700 | 1,400 | n.p. | 750 | 1,400 | n.p. | 600 |
| software | CUDA | CUDA | CUDA | ROCm | ROCm | ROCm | oneAPI |

† vendor figures, dense tensor peaks; Rubin from the September 2026 spec revision; MI455X compute in MXFP8 and MXFP4; PVC FP16 is Intel's figure, consistent with its confirmed TF32 rate; n.p.: not published as of November 2026; IC: Infinity Cache.

# Summary: what to take from this lecture

The dies converged, the platforms did not; the fabric, the software and the buying model decide.

**Eight things to remember**

1. On paper the compute GPUs converged: 10 PFLOP/s of FP4 from both AMD and NVIDIA in 2025; 35–40 in 2026.
2. AMD led on memory per GPU from 2021, tied in 2025.
3. FP64 diverged: AMD built it up (MI430X 288 TFLOP/s), NVIDIA cut it, Intel ran it at full rate once.
4. The scale-up fabric is NVIDIA's widest lead.
5. Packaging: Intel's 47 tiles, AMD's chiplets, NVIDIA's dies.
6. HPC is portable and buys FP64 per watt: the exascale list is AMD, Intel, one NVIDIA and one CPU-only machine.
7. AI is CUDA-first and buys by the quarter: NVIDIA's share follows from software and fabric.
8. Measure, do not read: MLPerf, HPL and microbenchmarks.

**Open questions for 2027**

**Does Helios deliver?** The first multi-gigawatt AMD deployments will show whether UALink over Ethernet and ROCm hold up at rack scale.

**Does Rubin's lead hold?** Rubin Ultra and the MI500 are both promised for 2027; both vendors' 2026 specs moved after announcement.

**Does Intel return?** The GPU D-series is an inference card on commodity memory; Jaguar Shores has no date; the NVIDIA deal makes Intel a CPU supplier to NVLink racks.

**What does HPC do?** FP64 at 288 TFLOP/s (AMD) against 33 (NVIDIA) with emulation; the 2028 machines will answer.

**Do the standards bite?** UALink silicon, Ultra Ethernet NICs and OCP racks in multi-vendor deployments, or NVLink Fusion.

# Sources

Whitepapers and datasheets, papers and talks, lists and reporting; † marks vendor material.

**AMD**†
CDNA 1, 2, 3 and 4 architecture whitepapers (2020–25); Instinct MI200, MI210, MI300A, MI300X, MI325X and MI355X datasheets and briefs; ROCm documentation (MI100, MI250, MI300 microarchitecture pages; release history); Smith and Alla, "MI300X generative AI accelerator and platform architecture," Hot Chips 36 (2024); AMD ISSCC 2024 MI300 paper; Hot Chips 37 CDNA 4 talk (2025); Advancing AI 2025 and 2026 announcements (MI400, MI455X, Helios, MI430X); Supermicro Helios datasheet (2026); AMD Q3 2026 results and call; AMD–OpenAI, AMD–Eviden and ORNL (Discovery, Lux) press releases; AMD MLPerf blogs (Inference 4.1, 5.0, 5.1, 6.0; Training 5.1).

**Intel**†
Junkins, "The compute architecture of Intel processor graphics Gen9" (2015); Gen11 architecture whitepaper (2019); Xe-HPG introduction and the oneAPI GPU optimization guide (Xe architecture); Data Center GPU Max 1550 specifications (ARK); Gomes et al., "Ponte Vecchio: a multi-tile 3D stacked processor for exascale computing," ISSCC 2022; Hot Chips 34 PVC talk (2022); Kaplan, "Gaudi 3," Hot Chips 2024; Intel newsroom (Max series, TOP500 November 2023); Intel Q3 2026 results; Intel–NVIDIA announcement (2025); ALCF Aurora documentation; Goto et al., "HPL and HPL-MxP at exascale on Aurora" (2026); Hatanpää et al., AERIS (2025); HPCwire, "Aurora the survivor" (2023); reporting on Rialto Bridge, Falcon Shores, Crescent Island and the GPU D-series.

**NVIDIA and platforms**†
HGX A100, HGX H100 and H200 datasheets; GB200 NVL72, GB300 NVL72 and Vera Rubin NVL72 product pages (2025–26); NVLink Fusion press release (2025); the previous lecture's sources for the architectures.

**Fabrics and standards**
UALink Consortium, UALink 1.0 white paper (2025); Ultra Ethernet Consortium 1.0 announcement (2025); OCP OAM and UBB specifications; Frontier architecture overview (OLCF, Abraham); LLNL El Capitan system page; Next Platform on El Capitan (2024); LUMI, CINES Adastra and HLRS Hunter pages.

**Software and portability**
Reguly, "Evaluating the performance portability of SYCL," SC-W 2023; Davis et al., "Taking GPU programming models to task for performance portability," ICS 2025; ORNL/TM-2025/4398, Kokkos with SYCL on AMD GPUs; Khronos SYCL 2020 release; Codeplay oneAPI plugins (2022); UXL Foundation (2023); SYCLomatic documentation; SemiAnalysis, "MI300X vs H100 vs H200: training" (2024).

**Lists and benchmarks**
TOP500, HPCG and Green500 lists, June 2026; HPE on the June 2026 list; MLCommons MLPerf Training 5.0 and 6.0, Inference 6.0; TrendForce (2026); market-share estimates (Silicon Analysts, 2026; secondary). Full list with links in the paper kit.